PROJECT BRIEF · SCRATCH (GR 6–8) INTERMEDIATE: DATA & SIMULATIONS

Lesson 9: Project: My Simulation

Module 3 | 50 minutes | Project: simulation | Grades 6–8 · ages 11–14

Name: _____ Class: _____ Date: _____

Your mission
Simulate a real-world process to answer a question: write rules and assumptions, build it with randomness and agents or time steps, measure a result, and run an experiment by changing one setting.

Checklist

- [ ] Question
- [ ] Rules and assumptions
- [ ] Randomness
- [ ] Agents or time steps
- [ ] Measurement
- [ ] Experiment and conclusion

Milestones

1. Question and rules — done on: _____
2. Smallest version — done on: _____
3. Measurement — done on: _____
4. Experiment — done on: _____
5. Conclusion and submit — done on: _____

Plan it

Sketch your stage and write what each sprite will do.

How it will be marked

Criterion	1 · Beginning	2 · Developing	3 · Secure	4 · Excellent
Model design	No rules	Vague rules	Clear rules and assumptions	Thoughtful rules with justified assumptions
Implementation	Broken	Partly works	Agents or steps with randomness work	Clean, well-structured simulation
Measurement	None	Shown but unclear	Clear measurement reported	Measured over time or across runs
Experiment	None	Changed several things	Changed one thing and compared	Repeated runs and fair comparison
Conclusion	None	Describes output	Answers the question	Answers and evaluates the model's limits

Want more?

- Record a list over time
- Two kinds of agent
- Sliders via key controls

Two stars and a wish

Ask a friend to test your project.

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